

EDUCATION

- **National School Of Computer Science** Manouba, Tunisia
Computer Science Engineering Sep. 2024 – Present
- **Preparatory Institute For Engineering Studies** Monastir, Tunisia
National Entrance Examination to the Engineering Cycle Sep. 2022 – Jul. 2024

EXPERIENCE

- **Summer Intern, Robotics & Multi-Agent Systems** Sousse, Tunisia
Enova Robotics Jun. 2026 – Jul. 2026
 - **Multi-Robot Coordination System:** Developed a full autonomous navigation stack for outdoor multi-robot coordination under ROS2, integrating Nav2-based path planning with LLM-driven task dispatching.
 - **Agentic Task Architecture:** Designed a multi-agent MCP server architecture where each fleet operation is handled by a dedicated agent, using LangGraph to orchestrate inter-agent workflows and enable natural language-driven, dynamic mission orchestration across the robot fleet.
- **Summer Intern, Web Development** Sousse, Tunisia
Code & Craft Jun. 2025 – Aug. 2025
 - **Project Management Platform:** Developed a full-stack web application allowing managers to create projects, assign tasks, and query project data using natural language commands via a Model Context Protocol server, reducing task assignment from ~10 minutes to under 30 seconds.
 - **Automation Tools:** Built custom tools for task tracking, deadline monitoring, and progress reporting, enabling automated task assignment and real-time project insights.
 - **Full-Stack Implementation:** Implemented the frontend with Angular and backend APIs using Node.js.
- **Robotics Club, ENSI** Manouba, Tunisia
Aeronautics Manager Jun. 2025 – Present
 - **Drone Development:** Built the university's first drone from scratch using an ESP32-based flight controller, integrating motors, sensors, and wireless communication modules.
 - **Team Leadership:** Managed the aeronautics team, assigned tasks, supervised testing, and coordinated project deadlines for competitions.

PROJECTS

- **CropEye : Autonomous Drone Monitoring System:**
Designed and built an autonomous agricultural drone controlled by an AI language model through the Model Context Protocol, enabling natural language mission execution. Developed the agentic AI control layer using MAVLink for flight command execution, the ground station application, and a real-time telemetry dashboard for live mission monitoring.
- **Maze-Solving Robot : Floodfill Navigation Algorithm:**
Designed and implemented a maze-solving robot capable of autonomously exploring and navigating an unknown maze using the Floodfill algorithm, computing optimal paths in real time from continuously updated distance maps.

ACHIEVEMENTS

- **Competitive Programming:** Solved +200 problems on platforms like LeetCode and Codeforces. Participated in university-level contests and coding challenges, strengthening problem-solving, algorithms, and data structures skills.
- **Robotics Competitions:** Built the first drone at our university and secured 2nd place in a project competition with the Robotics Club thanks to this project.

SKILLS

- **Programming:** C++, C, Python, JavaScript
- **Robotics & Embedded:** ROS 2 , Gazebo ,MAVLink ,ArduPilot, PID Tuning, SITL
- **Web Development & AI Agents:** MCP, LangGraph, LangChain, Angular, HTML, CSS, Node.js
- **Tools:** Git, VS Code
- **Concepts:** Data Structures & Algorithms, OOP
- **Languages:** Arabic (Native), English (Advanced), French (Advanced)